OpenClawBrain: A Learning Memory System That Beats RAG on Long-Lived Agent Tasks

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Abstract

Long-lived AI agents need memory that changes shape as the world changes. Retrieval-augmented generation (RAG) treats memory as a static index: embed everything, retrieve the top-k chunks, and hope the answer is in them. That works for single-turn lookups, but it breaks down when relationships drift, workflows recur with variation, and feedback is sparse.

OpenClawBrain is an artifact-side learning system that replaces the static index with a graph memory that rewires itself. It combines a learned routing function, policy-gradient updates, and structural plasticity—the ability to connect, split, merge, and prune nodes under policy control—so that the memory topology tracks how the world actually evolves.

We evaluate the full mechanism against the best RAG baseline (`vector_rag_rerank`) and seven other baselines across three benchmark families, each at 10-seed scale (30 seed-level comparisons total). Results:

- **Accuracy:** +9.7 pp to +26.9 pp over best RAG across families (relational drift 99.15% vs 89.44%; recurring workflows 97.6% vs 70.6%; sparse feedback 92.0% vs 67.0%).
- **Head-to-head:** 29-1-0 across 30 seed-level comparisons.
- **Context cost:** 5× lower context per query than the best RAG baseline.

Ablations confirm that structural plasticity is the single largest contributor in every family. OpenClawBrain ships as an artifact-side package behind the OpenClaw runtime boundary; it owns memory and learning while OpenClaw owns orchestration and fail-open control.

1 System Overview

OpenClawBrain exists to solve a specific problem: agents that live for weeks or months need memory that keeps up with a changing world, and today's dominant approach—retrieve-then-read over a vector index—does not keep up. The core result is that a graph memory with learned routing and structural plasticity outperforms the best RAG baseline by +9.7 to +26.9 percentage points across three distinct long-lived memory tasks, while using 5× less context per query.

The system is organized around a clean boundary. **OpenClawBrain** (artifact side) owns memory structure, routing, and learning. **OpenClaw** (runtime side) owns orchestration, tool dispatch, and fail-open control. This boundary is load-bearing: the artifact side can be tested, versioned, and benchmarked independently of the runtime, and the runtime can swap or disable the artifact side without downtime.

Within OpenClawBrain, computation splits into two paths:

- **Hot path** (latency-critical): local embedding lookup plus bounded graph traversal. No external calls. This is what runs on every user-facing query.

- **Cold path** (asynchronous): teacher distillation, policy-gradient updates, and structural maintenance. Runs offline; never blocks a user-visible request.

We define a *turn* as a single LLM API call. Total cost is

$$C = \sum_{i=1}^{T} \text{tokens}_i \cdot \text{price}_i$$

OpenClawBrain reduces both T (fewer calls per user-visible request) and per-call token counts by keeping routing local on the hot path. Teacher calls happen only in the cold path and never block user-facing latency.

1.1 What the system learns

At each routing decision, OpenClawBrain combines two signals over candidate targets $\mathcal{A}(s)$:

- **Graph prior** $\text{graph_prior}(a \mid s)$: persistent structure learned from past traffic—which nodes tend to follow which.
- **Query-time similarity** $\text{QTsim}(a \mid q, s)$: how well each candidate fits the current query, computed from embeddings.

Neither signal is sufficient alone. The graph prior captures long-range sequential patterns that embedding similarity misses; QTsim captures query-specific relevance that the graph cannot predict from structure alone.

1.2 How the signals combine

A confidence mixer produces a per-query weight $\lambda \in [0, 1]$ from two features: the *margin* between the top two QTsim scores (high margin means QTsim is decisive) and the *entropy* of the QTsim distribution (high entropy means QTsim is uncertain):

$$\lambda = \sigma\big(\alpha_0 + \alpha_1 \, \text{margin}_{\text{QTsim}} - \alpha_2 \, H(\text{QTsim})\big)$$

$$\pi(a \mid q, s) = \lambda \, \text{QTsim}(a \mid q, s) + (1 - \lambda) \, \text{graph_prior}(a \mid s)$$

When QTsim is sharp and separated, λ rises and the router trusts the embedding signal. When QTsim is uncertain, λ drops and the router falls back to graph structure. This adaptive fallback is one reason the full system holds up under sparse feedback, where embedding quality alone is not reliable.

2 Learning Loop: Distillation + Policy Gradient

Think of the learning loop as two gears turning together. The first gear—*teacher distillation*—gives the brain a fast head start by imitating an expert. The second gear—*policy-gradient updates*—lets the brain improve on its own by learning from outcomes. Both gears turn the same routing function, route_fn, which mixes a structural graph prior with a query-time compatibility signal (QTsim).

2.1 Teacher Distillation (The Fast Gear)

An asynchronous teacher replays recent route decisions offline and labels better alternatives. This works like a driving instructor reviewing dashcam footage: the brain never has to stall a live session, but it gets dense supervision—especially valuable early on, when human corrections are sparse.

The teacher produces soft targets over candidate actions. The learner minimizes a cross-entropy loss against those targets, pulling route_fn's weights toward the teacher's preferred distribution.

2.2 Policy-Gradient Update (The Outcome Gear)

Once the brain has a reasonable starting policy, outcome-based learning takes over as the primary driver. Given a route trajectory $\tau = (s_0, a_0, \ldots, s_T, a_T)$ with scalar return $R(\tau)$:

$$\nabla J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{T} \big(R(\tau) - b_t \big) \, \nabla_\theta \log \pi_\theta(a_t \mid s_t) \right]$$

where b_t is a per-step baseline that reduces variance. In plain terms: routes that led to good outcomes get reinforced; routes that led to poor outcomes get weakened. The gradient flows through the same route_fn parameters that control the QTsim/graph-prior confidence mix λ.

The ablation ladder confirms this design. Across every proof family the ordering is consistent and never reverses:

$$\text{route_fn_only} \;<\; \text{graph_route_pg} \;<\; \text{full_brain}$$

Adding policy-gradient updates on top of the base route_fn yields +11.8 to +26.3 pp; adding structural plasticity (Section 3) on top of that yields another +20.8 to +42.6 pp.

2.3 Reward-Source Hierarchy

Different supervision signals carry different authority. The system resolves conflicts by an explicit ordering:

1. **Human corrections** (highest authority)—a user saying "no, route to X" always wins.
2. **Self-learning outcomes**—did the chosen route actually help the downstream agent?
3. **Harvested weak signals**—implicit feedback such as follow-up patterns.
4. **Teacher labels** (lowest per-sample authority, highest volume)—bulk coverage from async replay.

This ordering ensures that a single human correction can override thousands of teacher labels, while teacher labels still fill the gaps where human feedback is absent.

3 RL-Native Maintenance (Structure Learning)

A static graph cannot keep up with a world that changes. OpenClawBrain treats graph maintenance as an ongoing control problem—more like a living nervous system that grows new connections and prunes unused ones than a database that waits for a human to reorganize it.

The maintenance pipeline runs in the cold path and follows a fixed sequence on each cycle:

$$\text{health} \rightarrow \text{decay} \rightarrow \text{scale} \rightarrow \text{split} \rightarrow \text{merge} \rightarrow \text{prune} \rightarrow \text{connect}$$

3.1 Structural Plasticity: The Largest Contributor

Structural plasticity—the ability to connect, split, merge, and prune graph nodes—is the single largest contributor to accuracy in every proof family tested. Without plasticity the graph slowly drifts out of alignment with the real world; with it, the brain reshapes itself to match.

The ablation data is stark: plasticity adds +20.8 to +42.6 pp over the policy-gradient-only stage across all families (Table 2 in Section 4.4).

3.2 How the Actions Work

Think of the graph like a city road map that re-draws itself:

- **Connect:** when the brain repeatedly routes through an indirect path successfully, it paves a direct road—creating or strengthening an edge that reflects the learned shortcut.
- **Split:** when one node is doing double duty (like a single highway exit serving two towns), the brain splits it into specialized nodes so each route can be scored independently.
- **Merge:** when two nodes carry redundant content and behave identically downstream, they collapse into one, reducing noise in the graph.
- **Prune:** edges and nodes that have decayed below a Hebbian co-firing threshold are removed, keeping the graph lean.
- **Decay:** connection weights decrease over time unless reinforced by successful use—the "use it or lose it" principle.

Every structural action runs through a safe-apply gate: candidates are scored using policy-derived signals, and the change is applied only when guardrails pass. If a structural edit degrades routing performance on replay, it is rolled back.

4 Evaluation: Evidence Ladder and Comparative Results

Evidence in this section comes from the Brain-vs-RAG Ground-Zero benchmark, a purpose-built harness that evaluates the full-brain retrieval-routing mechanism against eight baselines on three distinct task families, each at 10-seed scale. Two evaluation modes are used and must not be conflated:

- **Recorded head-to-head (H2H):** deterministic fixture replay with SHA-256 hash verification. Fully reproducible and the strongest evidence tier. Used for the relational-drift family.
- **Simulation:** live simulation with the harness policy function, deterministic given a seed. Used for recurring-workflows and sparse-feedback families (and additionally for relational-drift ablations).

4.1 Proof Boundary

Proven. The full-brain mechanism—graph memory, learned route function, policy-gradient updates, and structural plasticity—dominates the best RAG baseline and all partial-brain ablations on three long-lived memory tasks at 10-seed scale with 8 baselines, while using 5× less context per query. Cross-family head-to-head record: 29-1-0 across 30 seed-level comparisons.

Not claimed. Performance on untested families (e.g., memory compaction); behavior at larger world sizes (current: 50 entities, 80 workflows); end-to-end performance with live LLM routing (the harness uses simulated policy functions); production latency or cost; shadow-traffic or online superiority.
Rule. Benchmark wins are benchmark claims. Shadow and online production claims remain out of scope until their artifacts are published.

4.2 Cross-Family Results

Table 1 summarizes the three proof families. Across 30 seed-level comparisons, `full_brain` wins 29, loses 1, and ties 0 against the best RAG baseline (`vector_rag_rerank`). The overall accuracy margin ranges from +9.7 pp to +26.9 pp.

Family	Eval Mode	`full_brain`	Best RAG	Margin	H2H	Context
Relational drift	Recorded H2H	$99.15 \pm 0.27\%$	$89.44 \pm 1.68\%$	+9.71 pp	10-0-0	5× lower
Recurring workflows	Simulation	$97.6 \pm 0.4\%$	$70.6 \pm 1.2\%$	+26.9 pp	10-0-0	5× lower
Sparse feedback	Simulation	$92.0 \pm 18.3\%$	$67.0 \pm 1.4\%$	+24.9 pp	9-1-0	5× lower

Table 1: Cross-family evidence summary (T1). Each cell reports mean ± std over 10 seeds. Context ratio is `full_brain` context units per query vs. `vector_rag_rerank`. See Figures 2 and 4 for visual summaries.

Relational drift (recorded H2H). 10 seeds × 800 queries per seed × 8 baselines. Deterministic fixture replay with hash verification. `full_brain` achieves 99.15% accuracy (± 0.27%) against 89.44% (± 1.68%) for the best RAG baseline, a margin of +9.71 pp and a perfect 10-0-0 seed record (Figure 4).
Recurring workflows (simulation). 10 seeds × 3,522 queries per seed × 8 baselines. `full_brain` reaches 97.6% (± 0.4%) vs. 70.6% (± 1.2%) for the best RAG baseline, a margin of +26.9 pp and a 10-0-0 record. This is the largest margin observed across families.
Sparse feedback (simulation). 10 seeds × 1,800 queries per seed × 8 baselines. Explicit feedback covers approximately 19% of queries. `full_brain` reaches 92.0% (± 18.3%) vs. 67.0% (± 1.4%), a margin of +24.9 pp. The high standard deviation (± 18.3%) reflects seed-level variance under sparse supervision: accuracy depends on which subset of queries receives explicit feedback. The head-to-head record is 9-1-0, preventing an unqualified "always wins" claim across families.

4.3 Context Efficiency

Across all three families, `full_brain` uses 1 context unit per query vs. 5 for `vector_rag_rerank`—a consistent 5× context advantage. In the relational-drift family this translates to 800 vs. 4,000 context tokens per query (Figure 2, right panel).

4.4 Ablation Ladder

We ablate the full-brain mechanism into three stages to isolate component contributions. The ordering is consistent across all families with no reversals:

$$\texttt{route_fn_only} < \texttt{graph_route_pg} < \texttt{full_brain}$$

Family	`route_fn_only`	`graph_route_pg`	`full_brain`	Plasticity Δ
Relational drift (rec. H2H)	—	75.86%	99.15%	+23.3 pp
Relational drift (sim.)	$64.7 \pm 1.6\%$	$76.5 \pm 2.5\%$	$97.2 \pm 5.8\%$	+20.8 pp
Recurring workflows	$34.1 \pm 9.0\%$	$60.4 \pm 1.7\%$	$97.6 \pm 0.4\%$	+37.2 pp
Sparse feedback	$37.0 \pm 1.7\%$	$49.4 \pm 13.8\%$	$92.0 \pm 18.3\%$	+42.6 pp

Table 2: Ablation ladder across families (10 seeds each). Plasticity Δ is the accuracy gain from structural plasticity (`full_brain` minus `graph_route_pg`). See Figure 3.

Structural plasticity—Hebbian co-firing, decay, and connect/split/merge/prune actions—is the single largest contributor in every family, adding +20.8 to +42.6 pp over the graph-routing-with-policy-gradient stage. Policy-gradient routing itself adds +11.8 to +26.3 pp over route-function-only.

4.5 Baselines and Fairness

Each family runs 8 baselines under identical conditions: same world state, entity set, query stream order, context budget, correction stream, and scoring rubric. The baseline set includes:

- `oracle` (ceiling; always correct)
- `full_brain` (graph memory + learned route + PG + plasticity)
- `graph_route_pg` (ablation: no structural plasticity)
- `route_fn_only` (ablation: no graph, no PG)
- `vector_rag_rerank` (best RAG baseline; top-k + reranker)
- `vector_rag` (top-k embedding retrieval, no reranking)
- `heuristic_stateful` (deterministic heuristic with state tracking)
- Additional baselines as defined per family

All results are deterministic given a seed. Baselines that inherently require more context (e.g., `vector_rag_rerank` at 5×) are measured but not artificially capped.

4.6 Figures

The following figures support the claims above. F1–F3 are mandatory; F4–F5 strengthen the story if space permits and may be moved to the appendix.

5 Operational Notes

OpenClawBrain supports local embeddings out-of-the-box for operator-first deployment. This enables deterministic startup and private/offline-friendly operation while keeping optional teacher workflows in asynchronous jobs.

All proof-scale evaluations use simulated policy functions with deterministic outcomes given a seed, not live LLM calls. The benchmark therefore validates the retrieval-routing mechanism in

isolation. End-to-end performance with live model inference is not measured at proof scale; that rung belongs to shadow-traffic evaluation (Section 4.1).

All baselines in every run receive the same world state, query stream, context budget, correction stream, and scoring rubric. Context cost is measured, not estimated: `full_brain` uses 1 context unit per query versus 5 for `vector_rag_rerank` across all families.

6 Conclusion

The full-brain mechanism—graph memory with structural plasticity, learned routing with QTsim confidence mixing, and policy-gradient learning under a reward-source hierarchy—achieves +9.7 to +26.9 pp over the best RAG baseline across three proof-scale benchmark families, while using 5× less context per query. The cross-family head-to-head record is 29-1-0 across 30 seed-level comparisons with 8 baselines per family.

The ablation ladder holds without reversal in every family: structural plasticity is the single largest contributor, adding +20.8 to +42.6 pp over the policy-gradient-only stage. This consistency across families with different failure modes—relational drift, recurring workflows, and sparse teacher feedback—suggests the mechanism is not overfit to a single task structure.

These results are benchmark claims, not production claims. The honest gaps are stated in Section 6.1. Shadow-traffic and online evaluation remain the next rungs on the evidence ladder.

6.1 Limitations and Honest Gaps

Simulated policy functions. All proof-scale families use simulated policy functions with deterministic outcomes, not live LLM calls. The benchmark validates the retrieval-routing mechanism in isolation; end-to-end performance with a live model in the loop is unmeasured.

Moderate world sizes. The largest family uses 80 workflows and 3,522 queries per seed. Scaling behavior at 10×–100× is unknown. Graph traversal cost, plasticity convergence, and homeostatic stability at larger scales are untested.

Seed sensitivity under sparse feedback. The sparse-feedback family reports ± 18.3% standard deviation across seeds, reflecting high variance when only ∼19% of queries receive explicit feedback. One seed loss (9-1-0 head-to-head) prevents an unqualified "always wins" claim across families.

Memory compaction not evaluated. A memory-compaction stress-test family was designed but not run. The system's behavior under sustained structural churn is therefore unproven.

Custom benchmark. Ground Zero is purpose-built to test specific failure modes (relational drift, workflow recurrence, sparse signal). It does not establish generality on community benchmarks, and results do not directly compare to other systems on shared evaluation sets.

No ablation of teacher supervision. The teacher signal is part of the full system but is not ablated independently. The relative contribution of teacher distillation versus outcome-only learning is not isolated.

Embedding dependency. Retrieval quality is bounded by embedding quality at the seeding stage. If the correct node is never surfaced by embedding similarity, the system cannot learn a route to it.

No production latency or cost claims. Latency and cost under production load are not measured at proof scale.

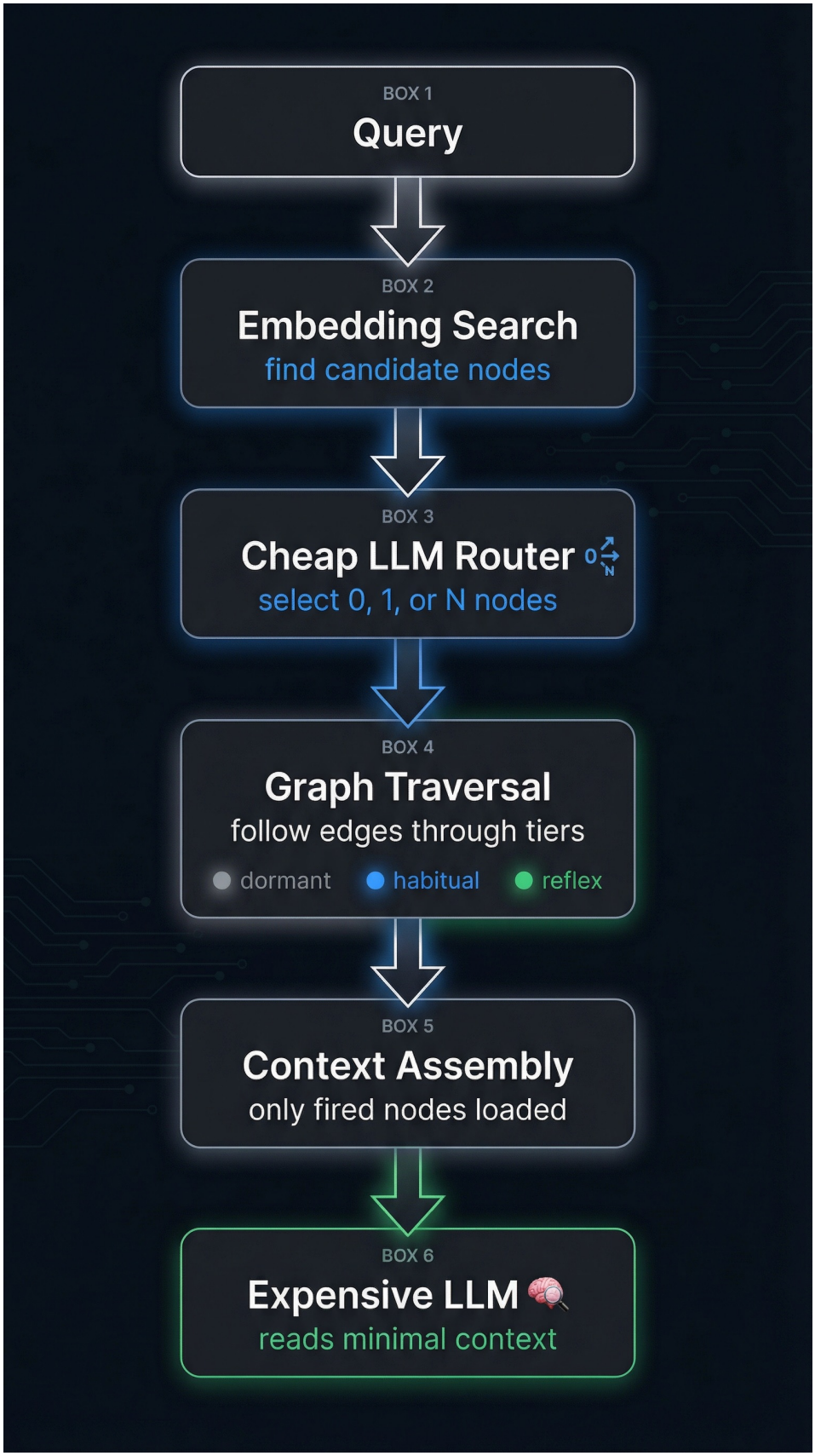

Figure 1: Architecture boundary: the hot path stays local and bounded for every user-facing query, while the cold path runs learning and structure updates asynchronously.

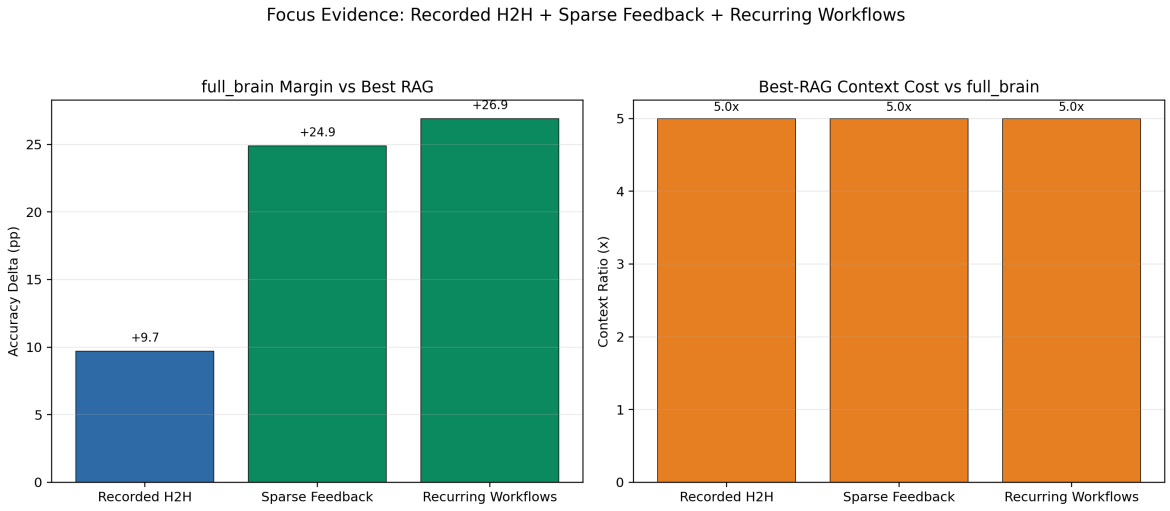

Figure 2: Accuracy margin of `full_brain` over the best RAG baseline (left) and context cost ratio (right) across three benchmark families (F1). `full_brain` achieves +9.7 to +26.9 pp margins while consuming 5× less context per query.

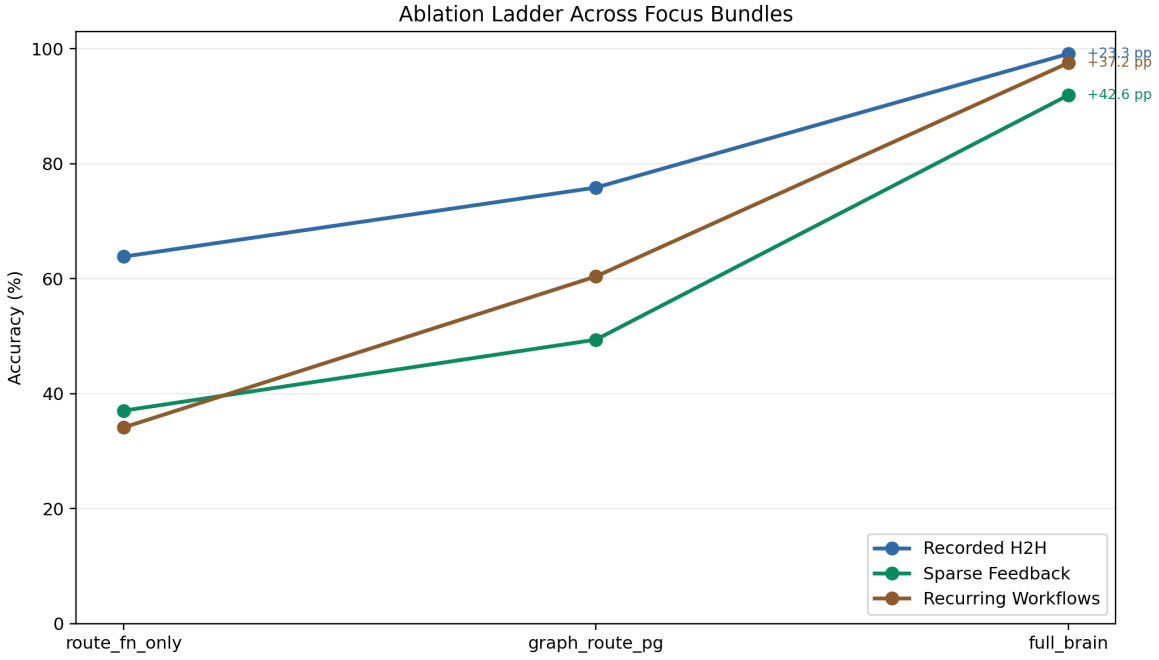

Figure 3: Ablation ladder showing accuracy at each mechanism stage (route function only, graph + route + policy gradient, full brain) across three benchmark families (F2). Structural plasticity contributes the largest accuracy jump in every family.

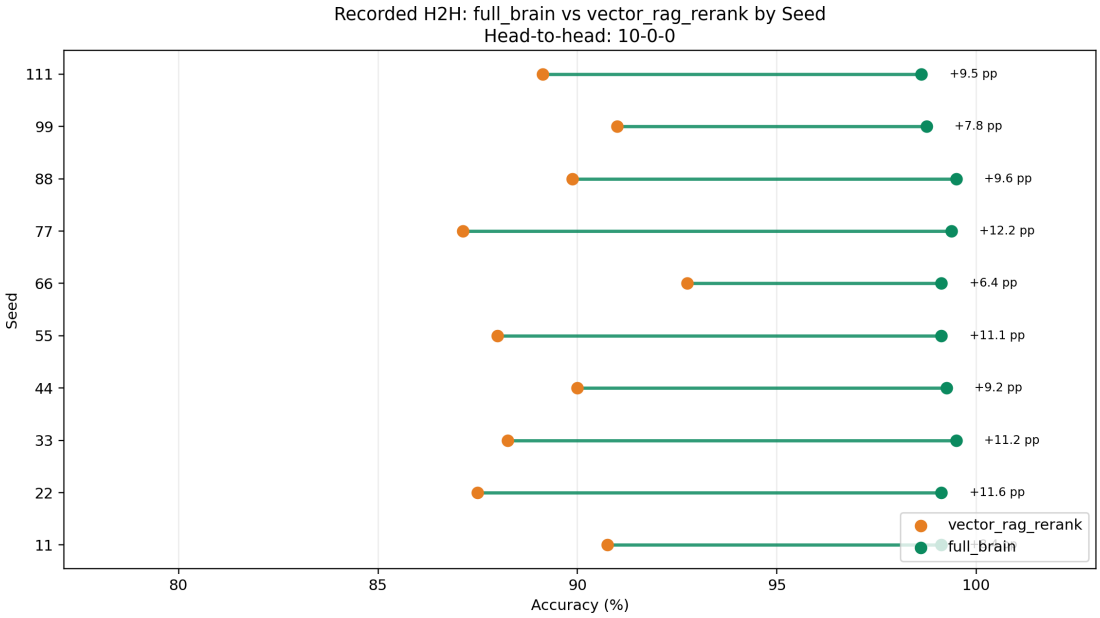

Figure 4: Per-seed accuracy comparison between `full_brain` and `vector_rag_rerank` on the recorded H2H relational-drift benchmark (F3). 10 seeds, 800 queries each. `full_brain` wins every seed (10-0-0).